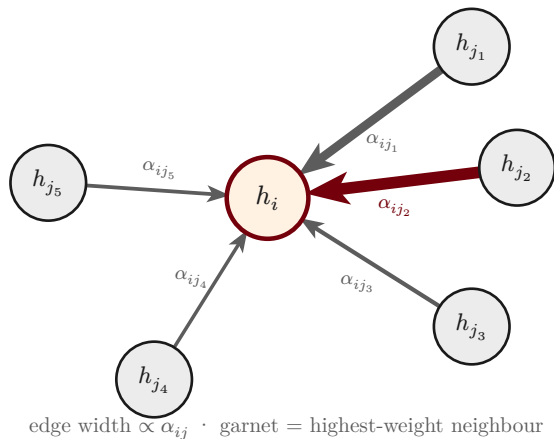


### (a) attention-weighted aggregation



### (b) computing $\alpha_{ij}$



linear map:  $z_k = Wh_k$

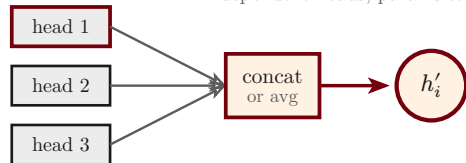
edge score:  $e_{ij} = \text{LeakyReLU}(a^\top [z_i \parallel z_j])$

**normalise (softmax over  $j \in \mathcal{N}(i)$ ):**

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

### (c) multi-head attention ( $K$ heads), then aggregate

$K$  independent heads, params  $W^{(k)}, a^{(k)}$



$$h'_i = \sigma \left( \sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right)$$

per head  $k$ , then concat  $\parallel_{k=1}^K$  (hidden) or average (output)